

ECE448 Spring 2026

L08-am-fm-

receivers



UNITED STATES
AIR FORCE
ACADEMY

SCHEDULE

Course schedule – see the ECE 448 syllabus.

TODAY

- FSK descriptions
- Broadcast FM (WBFM)
- Homework: Dual FM Receiver
- Narrow Band FM
- Homework: Rebroadcast

FM LAB

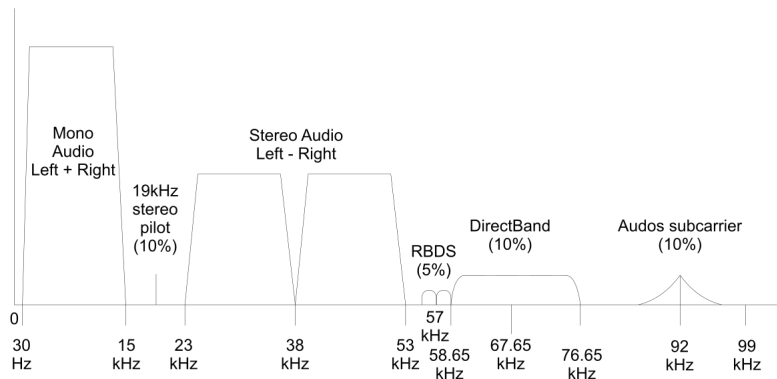
- $f_d = \underline{\hspace{2cm}}$
- $f_m = \underline{\hspace{2cm}}$
- Start with $a = \underline{\hspace{2cm}}$
- This means $\beta = \underline{\hspace{2cm}}$
- Now, create the flowgraph...

ELEMENTS OF THE FLOWGRAPH

- _____
- _____
- _____
- _____
- _____
- _____

- Now, create the flowgraph...

BROADCAST FM



DUAL FM RECEIVER

Assignment

Create a flowgraph that allows you to listen to two different FM stations at once.

Deliverables

- Audio of 2 FM stations
- Flowgraph (screenshot is ok)
- Report w/ description of flowgraph

Considerations

- Sample Rate/Bandwidth
- Frequency shift

NBFM

- File sink
-

TODAY

- FSK description
- FSK in GNURadio
- Parallel to Serial conversion (using byte data types)
- VCO block
- Save to file and view in audacity

TYPES OF MODULATION - AM, FM, PM

Amplitude Modulation

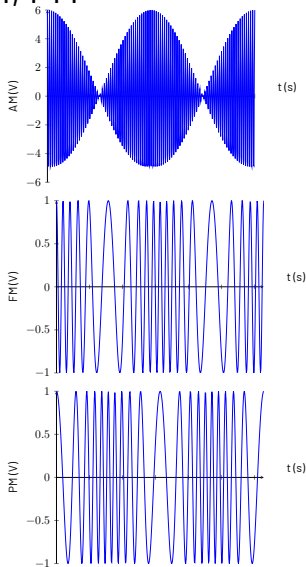
- Cheap, readily available technology
- Long distance transmission
- Uses less bandwidth

Frequency Modulation

- Noise Resistant
- Higher quality

Phase Modulation

- Noise Resistant
- Higher quality
- Used in cell phones



TYPES OF MODULATION - ANALOG VS. DIGITAL

- Modulating a digital signal is exactly the same as modulating an analog signal!
- **ANY** data can be modulated for transmission

